



Decoded EEG neurofeedback-guided cognitive reappraisal training for emotion regulation

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Received: 4 September 2023 / Revised: 6 March 2024 / Accepted: 17 March 2024 / Published online: 3 May 2024
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Abstract

Neurofeedback, when combined with cognitive reappraisal, offers promising potential for emotion regulation training. However, prior studies have predominantly relied on functional magnetic resonance imaging, which could impede its clinical feasibility. Furthermore, these studies have primarily focused on reducing negative emotions while overlooking the importance of enhancing positive emotions. In our current study, we developed a novel electroencephalogram (EEG) neurofeedback-guided cognitive reappraisal training protocol for emotion regulation. We recruited forty-two healthy subjects (20 females; 22.4 ± 2.2 years old) who were randomly assigned to either the neurofeedback group or the control group. We evaluated the efficacy of this newly proposed neurofeedback training approach in regulating emotions evoked by pictures with different valence levels (low positive and high negative). Initially, we trained an EEG-based emotion decoding model for each individual using offline data. During the training process, we calculated the subjects' real-time self-regulation performance based on the decoded emotional states and fed it back to the subjects as feedback signals. Our results indicate that the proposed decoded EEG neurofeedback-guided cognitive reappraisal training protocol significantly enhanced emotion regulation performance for stimuli with low positive valence. Additionally, wavelet energy and differential entropy features in the high-frequency band played a crucial role in emotion classification and were associated with neural plasticity changes induced by emotion regulation. These findings validate the beneficial effects of the proposed EEG neurofeedback protocol and offer insights into the neural mechanisms underlying its training effects. This novel decoded neurofeedback training protocol presents a promising cost-effective and non-invasive treatment technique for emotion-related mental disorders.

Keywords Decoded neurofeedback · Electroencephalogram (EEG) · Emotion classification · Cognitive reappraisal · Emotion regulation

Introduction

Neurofeedback is attracting renewed interest as a method wherein neural activity is measured and presented to the subject in real-time to facilitate self-regulation of the putative neural substrates underlying specific behavior of pathology (Sitaram et al. 2017). Typically, neurofeedback training approaches involve real-time recordings of brain activity using neuroimaging techniques, such as electroencephalography (EEG), functional magnetic resonance imaging (fMRI), and functional near-infrared spectroscopy

(fNIRS) (Sitaram et al. 2017; Patil et al. 2023; Chai et al. 2024). Emotion regulation stands out as one of the primary areas of research in neurofeedback training applications. Emotion regulation is crucial for maintaining both physical and mental health (Gross 2002) with difficulties in emotion regulation being implicated in the emergence and maintenance of psychopathology, such as depression (Gross and Jazaieri 2014; Donofry et al. 2016). Consequently, a key objective of psychotherapeutic interventions is to assist patients in improving their emotion regulation skills, and neurofeedback training emerges as a promising method for brain rehabilitation (Sloan et al. 2017).

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Conventional EEG neurofeedback

Previous studies have utilized regional brain activities to develop neurofeedback training protocols for emotion regulation. For instance, healthy individuals have demonstrated the ability to modulate their brain activity in response to real-time fMRI neurofeedback from various brain regions associated with emotion regulation, such as the amygdala, anterior insula, and anterior cingulate cortex (Linhartová et al. 2019). Compared to fMRI neurofeedback, EEG neurofeedback offers the advantages of being more widely accessible, cost-effective, and applicable in clinical settings. Existing EEG studies on emotion regulation have typically focused on single-frequency band EEG activity in specific brain regions for neurofeedback signal calculation, such as alpha symmetry (Quaedflieg et al. 2016; Mennella et al. 2017; Li et al. 2023), and high-beta down-training (Paquette et al. 2009; Cheon et al. 2016; Wang et al. 2019). However, the generation, perception, and regulation of emotions rely on the coordinated activity of multiple brain regions or networks (Morawetz et al. 2020). Conventional neurofeedback techniques that solely use regional brain activity as feedback may have limited efficacy in eliciting neurofeedback training effects on emotion regulation.

Cognitive reappraisal for emotion regulation

One of the most effective emotion regulation strategies is cognitive reappraisal, which involves consciously altering the interpretation of an event to modify its emotional impact (Gross 2015). Additionally, the neural correlates of cognitive reappraisal of emotions can be described using EEG (Gan et al. 2015; Aydın 2022, 2023). Empirical evidence consistently suggests that cognitive reappraisal is beneficial for psychological health in clinical settings (Forkmann et al. 2014). However, a particular challenge that may hinder the acquisition of cognitive reappraisal strategies through behavioral therapy is that individuals may lack metacognitive awareness of how psychotherapeutic strategies affect them. To address this issue, a previous study utilized real-time fMRI data collection and provided regional brain activities as neurofeedback during cognitive reappraisal training. Their findings suggested that the real-time feedback of regulation performance could assist subjects in learning to voluntarily self-regulate their brain signal in the brain circuitry of emotion regulation and facilitate the application of regulatory strategies to real-world situations in daily life (Zweerings et al. 2020). Additionally, both directions of regulation for cognitive reappraisal strategies, namely upregulating positive emotions and downregulating negative emotions, are crucial for controlling affective arousal (Gross 2015). However, thus far, the downregulation of

negative emotions has been the sole focus of previous neurofeedback studies (Linhartová et al. 2019; Wang et al. 2019; Zweerings et al. 2020), leaving a limited understanding of the potential benefits of neurofeedback for cognitive reappraisal training. Therefore, developing cognitive reappraisal training protocols based on EEG neurofeedback may offer more convenient neuroscience-based interventions for self-regulation training, and it is necessary to compare training performance between the two regulation directions.

Decoded EEG neurofeedback

With the assistance of machine learning techniques, emotional states can be decoded from human brain activity, and these decoded brain states can serve as a neurofeedback signal to make the setting of regulation targets more specific and objective (Shibata et al. 2019). While most previous decoded neurofeedback studies have been conducted using fMRI (Watanabe et al. 2017), Bu et al. developed a novel EEG neurofeedback protocol for addressing nicotine addiction based on similar principles, demonstrating both short- and long-term impacts on cigarette craving and smoking behavior. They achieved this by utilizing the decoded brain state related to smoking cue reactivity as the neurofeedback signal (Bu et al. 2019). Additionally, Huang et al. developed a neurofeedback training technique in which patients receive visual feedback on their current decoded emotional states to aid them in regulating their emotions toward a specific emotional state (Huang et al. 2021). These two studies verified the feasibility of developing a decoded neurofeedback protocol by collecting EEG signals and decoding brain states in real time.

Work in the current study

According to existing literature, neurofeedback technology, especially when combined with cognitive reappraisal for emotion regulation, shows promising clinical applications. However, previous studies mainly used fMRI, which may affect the feasibility of clinical implementation. Additionally, these studies have focused solely on reducing negative emotions, overlooking the importance of enhancing positive emotions. Therefore, in this study, we developed an EEG neurofeedback-guided cognitive reappraisal training protocol and assessed its effects on healthy subjects. Given the two regulatory directions for cognitive reappraisal, we utilized emotional stimuli with two valence levels. Specifically, each subject completed one neurofeedback training session comprising a calibration run and two emotional regulation runs for emotional stimuli with different valence levels (low positive and high negative). Initially, an individualized two-classes emotion classifier was trained using data collected

during the calibration run. Subsequently, during regulation runs, participants were instructed to employ cognitive reappraisal strategies to enhance the positive emotions or diminish negative emotions evoked by affective picture stimuli. Real-time feedback on their self-regulation performance was provided to aid them in strategy adjustment. This study employed a single-blind design and the enrolled subjects were randomly allocated to either the neurofeedback group with neurofeedback signal or the control group with no neurofeedback signal. We aimed to determine the neurofeedback training effect, compare outcomes between experimental groups, and investigate the changes in evoked brain activity underlying the brain's emotion-regulating processes.

Materials and methods

Subjects

Forty-two healthy subjects (20 females; 22.4 ± 2.2 years old) were recruited through an online advertisement at Shenzhen University. All subjects were right-handed and had normal or corrected-to-normal vision. None of the subjects had

prior experience with neurofeedback training. The study was approved by the Medical Ethics Committee of Shenzhen University (No. 2,019,017) and written informed consent was obtained from all subjects.

Emotional stimuli

This study utilized pictures to elicit emotions. Two hundred and forty color pictures were selected from the Chinese Affective Picture System (CAPS) datasets, comprising 120 positive pictures with high valence and 120 negative pictures with low valence, representing a variety of content as assessed by the reported norms for valence (Lu et al. 2005). Compared to negative pictures, normative valence ratings for positive pictures were significantly higher (positive: 6.62 ± 0.60 ; negative: 3.10 ± 0.94 ; $p < 0.001$), while both two groups of pictures were matched in terms of arousal ratings (positive: 5.92 ± 0.65 ; negative: 5.28 ± 0.44 ; $p < 0.001$).

Neurofeedback training

The experiment included one training session, comprising one calibration run and two regulation runs of neurofeedback training (as shown in Fig. 1). Throughout the

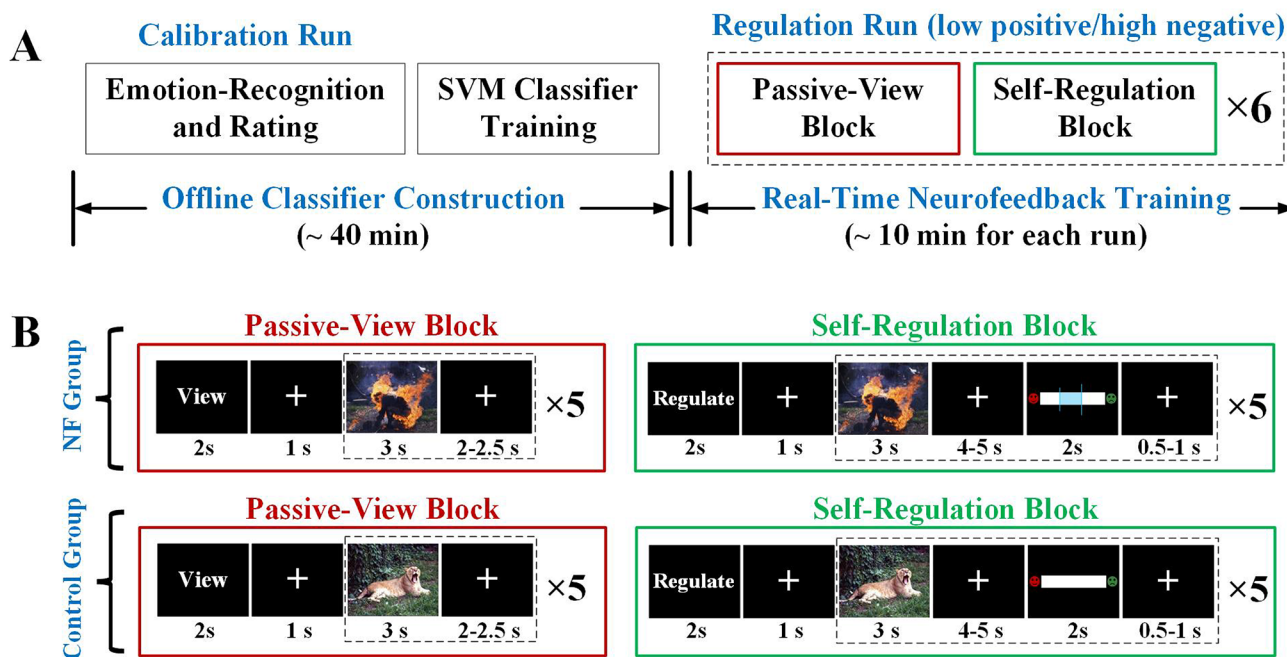


Fig. 1 Experimental procedure and neurofeedback paradigm. (A) The experimental procedure comprised two parts: one calibration run for constructing the offline emotion classifier and two neurofeedback training runs for emotion self-regulation using picture stimuli with varying emotional valence levels. Each calibration run included 120 trials of positive stimuli and 120 trials of negative stimuli, following reported norms of valence. Each regulation run comprised six pairs of passive-view blocks and self-regulation blocks, with the same pictures

presented in a different order. (B) During the neurofeedback training runs, each block consisted of five trials. For ‘view’ trials, subjects were instructed to respond naturally to each picture; for ‘regulation’ trials, subjects had to reappraise the pictures to increase the elicited positive affect or reduce the elicited negative affect. The neurofeedback signal, calculated as changes in the decoded emotional state, was displayed after each ‘regulation’ trial only for the neurofeedback (NF) group

experiment, subjects were seated in front of a computer monitor and instructed to remain as still as possible to minimize body movement artifacts. Subjects were randomly assigned to either the neurofeedback group (21 subjects) or the control group (21 subjects) in this single-blind study. Only the neurofeedback group received the feedback signal during the neurofeedback training.

Calibration run

In the calibration run, the subjects were presented with emotional stimuli of positive or negative valence (120 trials of each category), divided into three runs. Each trial began with the presentation of an image for 3000 ms, during which subjects were instructed to passively observe. Subsequently, they were asked to report their level of emotional valence on a 10-point scale (with a higher score indicating a higher level of positivity) within 1500 ms using the keyboard. The interstimulus interval (ISI) ranged from 500 to 1000 ms and was pseudorandomized within each run. Following the completion of the calibration run for each subject, a two-class linear support vector machine (SVM) classifier was trained using the recorded EEG data and subjective ratings, which were subsequently utilized in the following emotional regulation runs. Additionally, based on the subjective ratings of emotional valence recorded during the calibration run, 30 pictures with low positive valence (6.30 ± 0.39) and 30 pictures with high negative valence (0.98 ± 0.86) were selected for each subject.

Regulation run

After the calibration run, each subject underwent two neurofeedback training runs for emotional regulation using pictures with different levels of subjective-rated valence (LP for low positive, HN for high negative). Then two regulation runs were conducted for each set of selected pictures to compare the neurofeedback training performance. The order of the two regulation runs was counterbalanced.

Each regulation run comprised six pairs of blocks: one passive-view block followed by one self-regulation block, with each block containing five trials. Each pair of blocks featured the same five pictures but presented in a different order. Prior to the start of each block, a cue lasting 2000 ms was displayed on the screen, indicating either “passive-view” or “self-regulation”. During the “passive-view” block, each picture was presented for 3000 ms with an ISI of 2000 to 2500 ms. In the “self-regulation block”, following the presentation of the picture presentation and an ISI of 4000 to 5000 ms, the feedback signal was displayed for 2000 ms, reflecting the degree to which the brain activity matched the pattern for a positive emotional state. We implemented

intermittent feedback to reduce dual-task interference, as one recent study has suggested that intermittent feedback may be more effective in shaping brain activity compared to continuous feedback (Hellrung et al. 2018). The details of feedback signal calculation will be described in the [EEG data analysis](#) section. Following the neurofeedback signal, an additional ISI of 500–1000 ms was presented. For emotional self-regulation, subjects were instructed to employ cognitive reappraisal strategies to regulate elicited emotions with two distinct goals. All subjects received instructions on cognitive reappraisal strategies before the regulation runs (Goldin et al. 2008). During the LP regulation run, subjects were encouraged to enhance their positive emotional response through various cognitive reappraisal strategies, such as actively engaging with the situation, adopting a more positive reinterpretation of events, or perceiving the situation as more genuine and meaningful. Conversely, during the HN regulation run, subjects were instructed to reduce their negative emotional response through strategies such as reinterpreting the situation less negatively, observing it with detachment, cultivating psychological distance, or adopting a more objective perspective.

The affective state was assessed using the Positive and Negative Affect Scale (PANAS) (Schmukle et al. 2002) both before and after each regulation run. Additionally, a neurofeedback training effect questionnaire was used to measure subjects’ subjective feelings and strategies. The questionnaire included three questions: (i) please rate your level of fatigue after the completion of the experiment (1–9 points); (ii) please evaluate the level of concentration during training (scale 1 to 5); (iii) please assess the effectiveness of the strategy you used during training (scale 1 to 5); (iv) please briefly describe the cognitive reappraisal strategies employed during neurofeedback training.

EEG data acquisition

EEG signals were recorded using a BrainAmp amplifier (Brain Products GmbH, Germany) with 64 Ag/AgCl electrodes placed according to the extended international 10–20 system. Channels were recorded with a reference at FCz and a common ground at FPz. EEG signals were recorded using the BrainVision recorder software (Brain Products GmbH, Germany). The sampling rate was 1000 Hz for EEG data acquisition during calibration runs and resting-state runs, and 500 Hz for online acquisition during regulation runs. The impedance of all electrodes was kept below 10 k Ω .

EEG data analysis

The EEG data collected during the calibration run were analyzed offline before the regulation runs. During the

regulation run, the EEG signals were collected and analyzed in real time. Data analysis was conducted using EEGLAB (<https://sccn.ucsd.edu/eeglab>) and custom MATLAB (MathWorks, Inc.) scripts.

Preprocessing

The EEG signals from the calibration run were initially subjected to band-pass filtering (1–45 Hz) with a notch filter (49–51 Hz). The filtered data, which maintained the original sampling rate (500 Hz), formed Dataset (1). Additionally, the filtered data was downsampled to 125 Hz to generate Dataset (2). Then the continuous data (Datasets 1&2) were segmented into epochs (-500 to 3000 ms relative to stimulus onset), and baseline correction was performed using the pre-stimulus interval. Independent Component Analysis (ICA) was applied to remove artifacts from the data (Bell and Sejnowski 1995). To speed up the process of ICA decomposition, Dataset 2 was used for ICA calculation, and the resulting unmixing matrix was then applied to Dataset 1. Non-brain components in Dataset 1 were automatically identified and removed using IClab plugins, which detect artifacts such as eye movements and muscle artifacts (independent components with an 80% chance of being an artifact) (Pion-Tonachini et al. 2019). Finally, epochs of Dataset 1 were referenced to a common average and used for subsequent analysis.

For the passive view condition of the regulation run, data analysis was conducted upon completion of each block. Conversely, for the self-regulation condition of the regulation run, data analysis was conducted after completing each trial to calculate the feedback signal. It was assumed the brain and artifacts sources remained stable throughout the experiment, and the unmixing matrix obtained during the data analysis of the calibration run was utilized for online ICA correction during the regulation run. The remaining preprocessing steps were consistent with those described above for the calibration run.

EEG feature extraction

We extracted time-domain and frequency-domain features from the EEG signal of each epoch and channel, which have been widely used for classifying emotions in previous studies (Rahman et al. 2021). Hjorth parameters (activity, mobility, and complexity) were calculated from the non-linear analysis in the time domain (Li et al. 2018; Joshi and Ghongade 2021). We also extracted the amplitudes of N1 (130–180 ms), P1 (80–130 ms), N2 (180–280 ms), P2 (180–280 ms), P300 (280–400 ms), and LPP (400–500 ms) of event-related potentials (ERPs) in the time domain as features that are related to emotions (Goyal et al. 2015;

Ding et al. 2017; Singh and Singh 2021). Furthermore, we estimated power spectral density (PSD) using Welch's algorithm (1024-point FFT, 50% Hanning window), and extracted power features in five frequency bands: delta (1–4 Hz), theta (4–8 Hz), alpha (8–13 Hz), beta (13–30 Hz), and gamma (30–45 Hz) (Wang et al. 2014; Gao et al. 2020). The differential entropy (DE) was calculated for each frequency band, which is equivalent to the logarithmic PSD (Shi et al. 2013; Zheng et al. 2019; Patel et al. 2021). The discrete wavelet transform (DWT) was performed for feature extraction in the transform domain (Wang et al. 2014; Mohammadi et al. 2017). DWT analyzes signals in various frequency bands at different resolutions by decomposing the signal into coarse approximations and detailed information. In this study, we utilized the Daubechies 4th order due to its favorable time-frequency properties and decomposed the signal into four levels: D4 (gamma), D5 (beta), D6 (alpha), and D7 (theta). After concatenating EEG features independently obtained from 63 EEG channels to form a single feature vector, the final dimensionality is 1701 for each data segment.

Emotion classification model

The features extracted from the calibration run were used to train subject-specific SVM models for emotion classification. These models were then employed for online prediction during the regulation runs. To enhance the efficiency of the classification process, we implemented feature selection using the Eigenvector Centrality Feature Selection (ECFS) algorithm (Roffo and Melzi 2016). This algorithm constructs a feature graph, where each feature is represented as a node and evaluates the significance of each node through an indicator of centrality, i.e., eigenvector centrality. We applied the linear SVM algorithm and ten-fold cross-validation to compare the classification performance of the feature subsets. The top k (where $k=6, 11, 16, \dots, 1701$) features with the best classification accuracy were selected as the optimal feature subset. With each iteration, every feature was normalized to the subjects in the training sample by subtracting the minimum and dividing by the difference of the maximum minus the minimum, and the same maximum and minimum values were used to scale the test data. Subsequently, based on the determined value of k , hyperparameter optimization was performed for the ECFS algorithm. The hyperparameter, denoted as a , was initialized at 0 and incremented by a step size of 0.01 until reaching a value of 1. Classifier construction under ten-fold cross-validation was carried out for each value of a , and the value of a resulting in the highest average decoding rate was determined. Finally, using the determined values for k and a , the final prediction accuracy was evaluated using ten-fold cross-validation.

This process resulted in the creation of one trained classifier model for each subject.

Online emotion prediction and feedback calculation

During the regulation runs, EEG data were first preprocessed online. For paired trials (one from the passive-view block, and the other one from the self-regulation block) with the same stimulus picture, features were extracted (top k features with the best classification accuracy) and fed into the trained classifier from the calibration run separately. The classifier estimated probabilistic scores ranging from 0 to 1 in real time. When the current brain activity pattern was more similar to the positive emotional state, the score increased. The calculated probability scores for the same stimulus picture in both the passive-view block (short line) and self-regulation block (long line) were then displayed as horizontal bars (as shown in Fig. 1B). Thus, subjects in the neurofeedback group were informed of their self-regulation performance regarding the evoked emotion by comparing the position of the probability scores. Subsequently, the difference between the probability scores was recorded as the feedback signal value. Positive values indicated an increase in valence, while negative values indicated a decrease in valence. Subjects were instructed to enhance the change of probability scores in the positive direction by adjusting their cognitive reappraisal strategies.

Evaluation of neurofeedback training performance

The index of neurofeedback training performance during the emotion regulation run was calculated as the mean of the neurofeedback signals, which represented the differences of decoded emotional state between the passive-view and self-regulation blocks, across all trials in the self-regulation block. This index could reflect the overall amount of change in the emotional state due to self-regulation.

EEG feature analysis

We explored the selected EEG features for subject-dependent emotion classification from a group-level perspective by combining data from the neurofeedback group and the control group. Specifically, for each feature at each channel location, we calculated the percentage of subjects who had selected the current feature among the total number of subjects in both groups. This percentage indicated the degree of robustness of the feature at the group level. Features with a percentage value greater than the average were selected as discriminating features for emotion classification and were further analyzed. Seven Regions of Interest (ROIs) were selected based on previous literature

(Wang et al. 2014; Zheng and Lu 2015; Zhuang et al. 2018; Zheng et al. 2019): the left anterior frontal ROI1 (including Fp1, AF7, and AF3), the right anterior frontal ROI2 (including Fp2, AF8, and AF4), left temporal ROI3 (including FT9, FT7, T7, TP9, and TP7), right temporal ROI4 (including FT10, FT8, T8, TP10, and TP8), frontal Central ROI7 (including FC1, FC2, CZ, and Fz), left Parietal ROI5 (including P5, P7, and PO7), and right parietal ROI6 (including P6, P8, and PO8) (as shown in Fig. 4C). For the group-level discriminating features, each category of selected EEG features in each ROI region was averaged, and summarized feature values of each ROI region were calculated.

Statistical analysis

The chi-square test was used for gender, and independent two-sample t-tests were used for age and psychological ratings of behavioral scales, which were rated only once. Mixed effects ANOVA (between-subject factor: neurofeedback group vs. control group; within-subject factor: before vs. after training) was conducted on the PANAS ratings separated for the positive and the negative subscale. Additionally, a mixed two-way ANOVA was performed to compare subjective ratings of the emotional valence of picture stimuli used in regulation runs and the calculated index of neurofeedback training performance. Post-hoc tests with Bonferroni's correction were applied to examine significant main effects. The permutation tests ($N=1000$) were applied to assess the statistical significance of the emotion classification accuracy (Combrisson and Jerbi 2015), while a two-sample t-test was performed to compare the prediction accuracy of the emotion classification model constructed using EEG data from the calibration run between neurofeedback and control groups. Pearson's correlation was used for correlation analysis between classification accuracy and neurofeedback training performance for each group.

Average values of the selected EEG features, averaged within seven ROIs, were initially calculated by averaging all the view trials and regulation trials separately for each training run. Paired t-tests were then performed to compare the EEG features of the two groups' seven ROIs before and after emotion regulation. Then the amount of change in the features before and after emotion regulation was calculated separately for the two groups based on the significantly different EEG features. Bonferroni correction was applied to counteract the multiple comparisons among ROIs ($p < 0.05/7$). For ROIs with significant differences, Pearson's correlation analysis was performed between the changes in EEG features and the neurofeedback training performance.

Table 1 Demographics and behavioral ratings

Variable	NF Group (<i>n</i> = 21)	Control Group (<i>n</i> = 21)	<i>p</i> -value
Age, years	22.9 (1.9)	21.8 (2.4)	0.10 ^a
Education, years	15.0 (1.7)	14.3 (2.0)	0.20 ^a
Gender, male/female	10/11	12/9	0.76 ^b
PHQ-9	11.6 (2.1)	12.3 (3.4)	0.42 ^b
GAD-7	7.9 (1.5)	8.3 (2.8)	0.58 ^b
ERQ, reappraisal	27.3 (5.5)	28.5 (4.3)	0.44 ^b
ERQ, suppression	15.2 (2.9)	13.9 (5.2)	0.33 ^b
PANAS positive, pre	22.7 (6.8)	23.0 (7.8)	—
PANAS positive, post	30.2 (4.0)	32.2 (8.0)	—
PANAS negative, pre	18.8 (6.8)	17.2 (7.1)	—
PANAS negative, post	28.9 (5.5)	29.8 (8.2)	—
Self-reported fatigue score	5.1 (2.0)	5.8 (1.8)	0.29 ^b
Self-reported attention score	4.1 (0.6)	3.9 (0.7)	0.19 ^b
Self-reported validity score	3.3 (0.7)	3.6 (0.7)	0.19 ^b
Emotional valence, stimuli of low positive	6.27 (0.37)	6.32 (0.41)	0.67 ^b
Emotional valence, stimuli of high negative	0.92 (0.73)	1.04 (0.98)	0.67 ^b

Values are mean and values in parentheses are 1 standard deviation (SD).

NF = Neurofeedback; PHQ-9 = Patient Health Questionnaire-9; GAD-7 = Generalized Anxiety Disorder-7; ERQ = Emotion Regulation Questionnaire; PANAS = Positive and Negative Affect Scales.

a: two-sample *t*-test; b: Chi-square *t*-test.

Results

Demographic and behavioral ratings

The neurofeedback group and control group did not differ significantly with respect to gender, age, PHQ-9 ratings, GAD-7 ratings, and ERQ ratings (all $p > 0.05$). Mixed ANOVA using the pre- and post-test scores of PANAS, revealed no significant main effect of group (positive subscale, $p = 0.466$; negative subscale, $p = 0.750$), nor a significant interaction effect (positive subscale, $p = 0.330$; negative subscale, $p = 0.197$). However, a significant effect of training was observed (positive subscale, $p = 0.002$; negative

subscale, $p < 0.001$), indicating increased positive and negative effects after training for both groups. No significant group difference was observed for the self-reported subjective feeling of fatigue, concentration of attention, and neurofeedback training success (all $p > 0.05$) (Table 1).

Neurofeedback training performance

The neurofeedback training performance, indicative of changes in emotional states decoded from brain signals, was assessed across both experimental groups. The index of training performance, calculated for each subject, reflected the overall alternation in emotional state. A mixed two-way ANOVA conducted on the performance index revealed a significant interaction effect between experimental grouping and regulation type ($p = 0.005$). However, there were no significant main effects observed for group ($p = 0.361$) and regulation type ($p = 0.091$). Further analysis with post-hoc tests using Bonferroni-adjusted significance levels indicated a significant difference in neurofeedback training performance between the two groups during the LP regulation run ($p = 0.031$). In contrast, no significant difference was observed for the HN regulation run ($p = 0.419$) (Fig. 2).

Emotion classification accuracy and emotion-predictive features

The accuracies of the constructed emotion decoders for online prediction during the regulation run are shown in Fig. 3. The classifiers were trained using data collected from the emotion recognition and rating task, and the accuracy exceeded 50% for all subjects. Specifically, the mean accuracy for the neurofeedback group was 68% (range: 62–74%), and the mean accuracy for the control group was 68% (range: 58–78%). There was no significant difference in the mean classification accuracy of the pre-neurofeedback offline classifier between the two groups ($p = 0.820$) (as shown in Fig. 3A). The average confusion matrices are shown in Fig. 3B. Notably, no significant correlation was

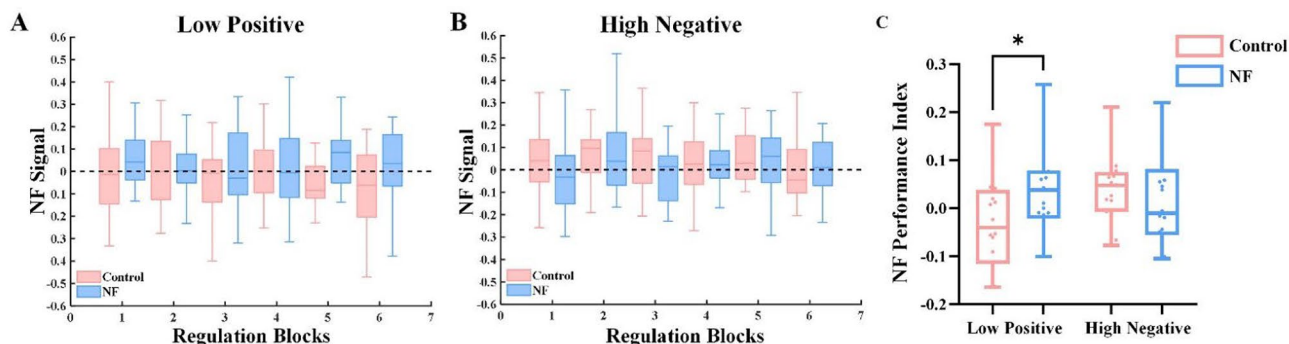


Fig. 2 Neurofeedback training performance of emotion regulation. * $p < 0.05$; error bar (SD); NF: neurofeedback

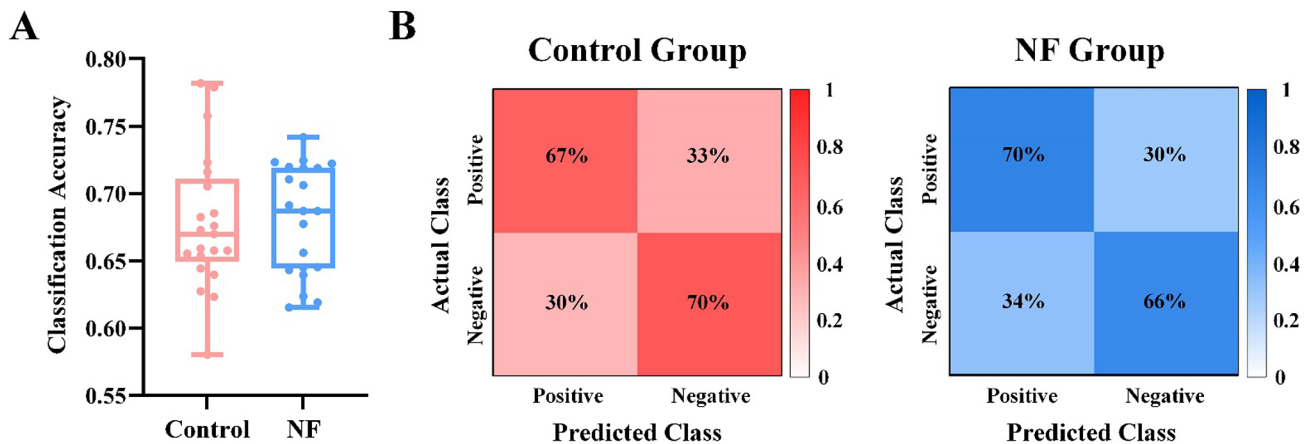


Fig. 3 Classification performance of the constructed emotion decoder. (A) Comparison between the neurofeedback (NF) group and control group in terms of classification accuracy. (B) Averaged confusion matrices of each group

observed between classification accuracy and neurofeedback training performance.

Fig. 4 presents the results of the group-level analysis of the selected informative EEG features used for constructing the emotion classifier. The analysis revealed that certain ERP features, such as P1 and LPP recorded from occipital-parietal regions, as well as N2, P2, and P300 recorded from parietal and central-frontal regions, were among the top-selected features. Additionally, Hjorth parameter features, specifically the complexity and mobility, measured from bilateral temporal and anterior frontal electrodes, were frequently selected. Regarding PSD features across different frequency bands, beta and gamma-band power features from anterior frontal electrodes were consistently chosen. Similarly, DE features, primarily from the high-frequency bands, are also prominent and are located in the anterior frontal, bilateral temporal, and parietal-occipital regions. Among the wavelet features, the energy of beta and gamma in the anterior frontal regions was consistently selected at a higher frequency.

Neural correlates of neurofeedback training

While neurofeedback training altered the emotional states and exhibited significant differences in regulatory effects between the two groups, we also explored changes in related EEG correlates during the LP regulation run. Therefore, paired sample t-tests were conducted on the selected EEG features averaged within seven ROIs to compare the view block and regulation block (refer to Table 2). Our findings indicate that the neurofeedback group showed significant changes in ERP potentials in the middle frontal (P300) and parietal regions (P1 and P2), Hjorth mobility in the anterior frontal regions, DE features in the anterior frontal regions (low-frequency rhythm: delta, theta, and alpha), and

high-frequency Wavelet energy and entropy in the frontal and temporal regions (Beta and Gamma). The control group showed significant changes in ERP potentials in the middle frontal (P300) and parietal regions (N2 and P2), Hjorth mobility and complexity in the anterior frontal and right temporal regions, DE features in the anterior frontal regions (low-frequency rhythm: delta and theta), and high-frequency Wavelet energy of the frontal regions (Beta).

Significant correlations were observed between neurofeedback training performance and regulation-related changes in brain activity among the selected EEG features in the LP regulation run (Fig. 5). Specifically, in the neurofeedback group, a significant correlation was found between training performance and the degree of change in high-frequency Wavelet energy in the left temporal region ($r=0.616$, $p=0.003$). Additionally, significant correlations were noted for the control group in the Hjorth mobility of the anterior frontal region ($r=0.576$, $p=0.006$).

Discussion

Neurofeedback training performance for emotion regulation

Neurofeedback is an innovative tool that has been successfully utilized to improve emotion control in recent years (Linhartová et al. 2019; Melnikov 2021). The utilization of EEG neurofeedback for enhancing emotion regulation holds the potential to augment the training effect through customized, targeted protocols, particularly considering its relative affordability, mobility, and ease of use compared to fMRI. For instance, frontal alpha asymmetry EEG neurofeedback has been employed in both healthy individuals (Quaedflieg et al. 2016; Li et al. 2023) and depressed subjects for

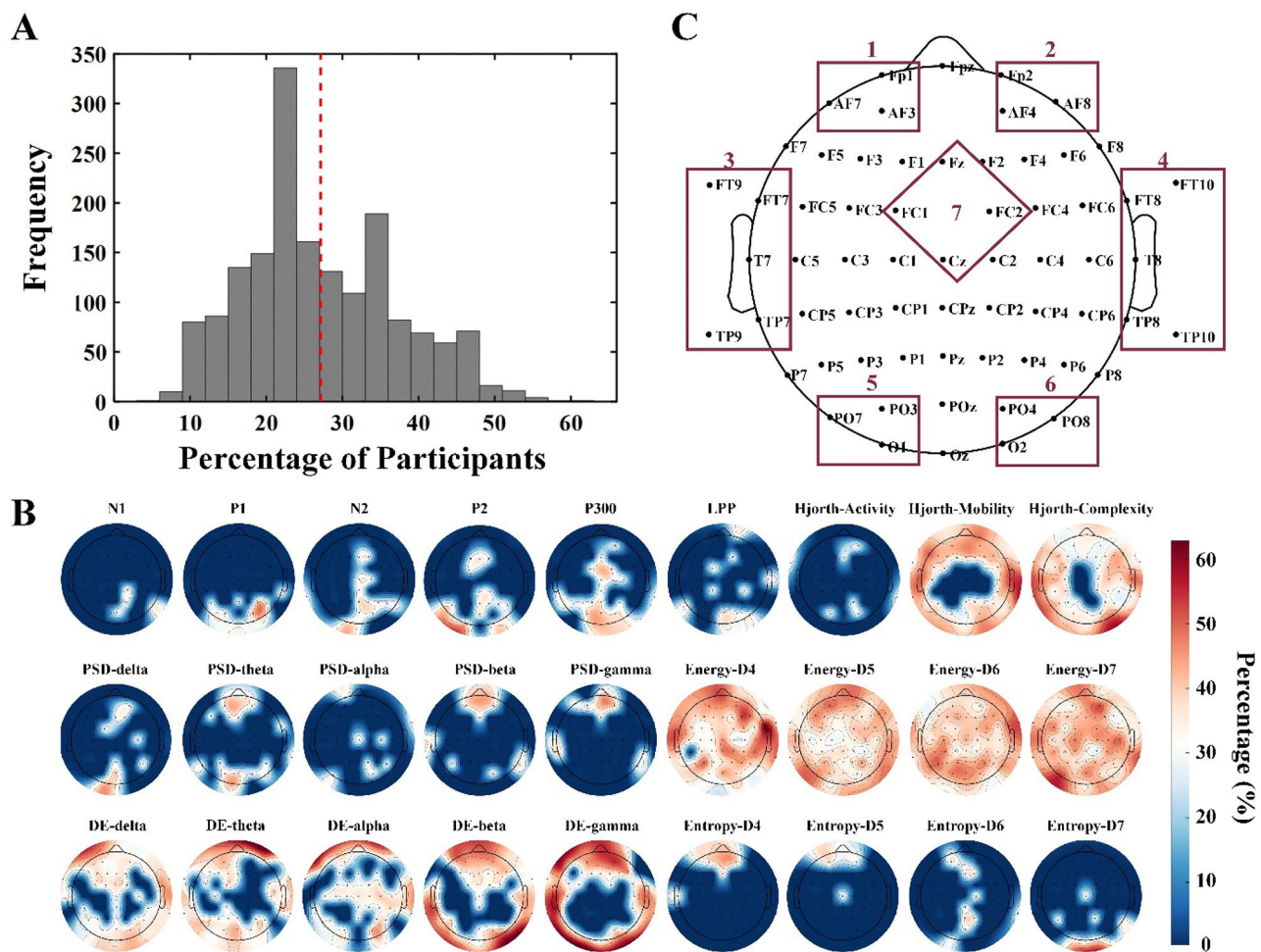


Fig. 4 The group-level analysis results of the selected EEG features for the construction of the emotion classifier. **A**: The percentage of subjects who had each feature selected and used for emotion decoder construction. **B**: Topographic figures showing the features that were

selected by more subjects (>27%, group mean) at the group level. **C**: Regions-of-interest (ROIs) for subsequent neural correlates analyses of neurofeedback training

emotion regulation training (Wang et al. 2019). A recent study introduced a real-time EEG-based brain-computer interface system for neurofeedback training (Huang et al. 2021). In their setup, the decoded instantaneous emotional state was provided as a neurofeedback signal to assist subjects in updating their strategies to regulate their emotions toward a specific emotional state suggested by the displayed video clips. Numerous studies have demonstrated that cognitive reappraisal stands out as one of the most versatile, effective, and adaptive regulatory strategies for modulating subjective emotional experiences (Webb et al. 2012). Less sadness, fewer negative consequences, and more life satisfaction are linked to a larger dependence on adaptive and efficient cognitive reappraisal strategies for emotional regulation. However, existing studies of EEG neurofeedback do not involve the cognitive reappraisal of emotionally

evocative stimuli. Therefore, it is necessary to develop an EEG neurofeedback paradigm that specifically targets cognitive reappraisal training for emotion regulation.

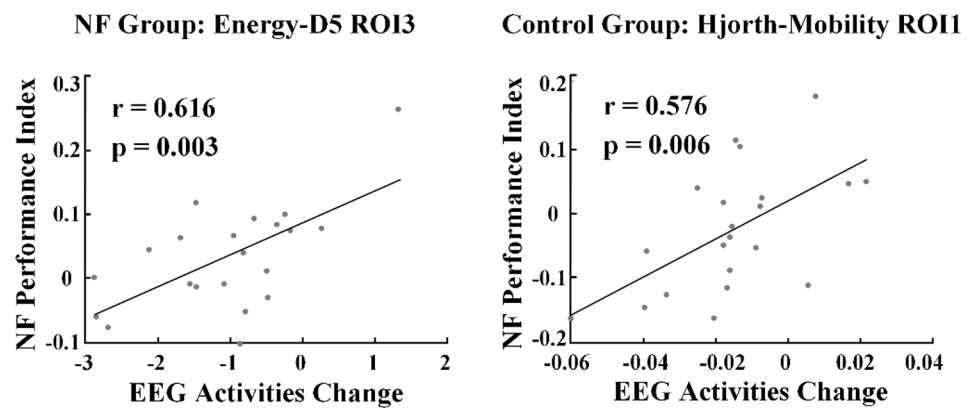
In this study, we constructed and evaluated a novel EEG neurofeedback-guided cognitive reappraisal training protocol. Cognitive reappraisal strategies involve two regulatory directions: upregulating positive emotions and downregulating negative emotions, both crucial for effectively managing affective arousal (Gross 2015). These strategies are associated with distinct neural mechanisms, as evidenced by fMRI studies (Sokołowski et al. 2022). Previous neurofeedback studies on emotion regulation have predominantly concentrated on the downregulation of negative emotions (Linhartová et al. 2019; Wang et al. 2019; Zweerings et al. 2020), with cognitive reappraisal being combined with only one fMRI neurofeedback training (Zweerings et al. 2020).

Table 2 Comparison of discriminating features for emotion classification between the view block and regulation block in the neurofeedback (NF) group and the control group

	ROI1		ROI2		ROI3		ROI4		ROI5		ROI6		ROI7	
	NF	Control	NF	Control	NF	Control	NF	Control	NF	Control	NF	Control	NF	Control
N1	\	\	\	\	\	\	\	\	\	\	\	\	\	\
P1	\	\	\	\	\	\	\	\	-4.15**	\	\	\	\	\
N2	\	\	\	\	\	\	\	\	\	4.14**	\	4.14**	\	-4.23**
P2	\	\	\	\	\	\	\	\	\	4.30**	3.11	4.04*	\	-5.16**
P300	\	\	\	\	\	\	\	\	\	\	\	\	-3.93*	-4.08*
LPP	\	\	\	\	\	\	\	\	\	\	\	\	\	\
Hjorth-Activity	\	\	\	\	\	\	\	\	\	\	\	\	\	\
Hjorth-Mobility	3.85*	3.55*	3.03	4.14**	\	\	\	3.63*	\	\	\	\	\	\
Hjorth-Complexity	\	\	\	3.95*	\	\	\	\	\	\	\	\	\	\
PSD-delta	\	\	\	\	\	\	\	\	\	\	\	\	\	\
PSD-theta	\	\	-3.14	\	\	\	\	\	\	\	\	\	\	\
PSD-alpha	\	\	\	\	\	\	\	\	\	\	\	\	\	\
PSD-beta	\	\	\	\	\	\	\	\	\	\	\	\	\	\
PSD-gamma	\	\	\	\	\	\	\	\	\	\	\	\	\	\
DE-delta	-4.20**	-3.65*	-4.20*	-3.85*	\	\	-3.07	\	\	\	\	\	\	\
DE-theta	-4.39**	\	-4.52**	-4.04*	\	\	\	\	\	\	\	-3.57*	\	\
DE-alpha	\	\	-3.27*	\	\	\	\	\	\	\	\	\	\	\
DE-beta	\	\	-3.26*	\	\	\	\	\	\	\	\	\	\	\
DE-gamma	\	\	-3.27*	\	\	\	\	\	\	\	\	\	\	\
Energy-D4	-2.10	\	\	\	\	\	\	\	\	\	\	\	\	\
Energy-D5	3.14*	3.27*	3.96**	\	4.47**	\	3.11	\	\	\	\	\	\	\
Energy-D6	\	\	\	\	\	\	\	\	\	\	\	\	\	\
Energy-D7	\	\	\	\	\	\	\	\	\	\	\	\	\	\
Entropy-D4	\	\	\	\	\	\	\	\	\	\	\	\	\	\
Entropy-D5	\	\	\	\	\	\	\	\	\	\	\	\	\	\
Entropy-D6	3.91*	\	\	\	\	\	\	\	\	\	\	\	\	\
Entropy-D7	\	\	\	\	\	\	\	\	\	\	\	\	\	\

Note: Only T values with $p < 0.05/7$ are shown in the table; * $p < 0.005$, ** $p < 0.0005$. “\” means no significant result.

Fig. 5 Results of the correlation analysis between the amount of EEG activity changes before and after the neurofeedback and neurofeedback training performance in the neurofeedback (NF) group and the control group



Consequently, existing research fails to provide a comprehensive understanding of whether neurofeedback can effectively aid in cognitive reappraisal training. Moreover, this study utilized fMRI signal changes between reappraisal regulation and viewing conditions as the neurofeedback signal for voluntary control of brain activation in key emotion regulation areas (Zweerings et al. 2020). This type of neurofeedback signal could not directly reflect cognitive reappraisal performance. Based on this consideration, we conducted real-time emotion classification, calculated the change in the decoded brain state between reappraisal regulation and viewing conditions, and utilized it as the neurofeedback signal to guide self-regulation performance. Our findings indicate that the neurofeedback group exhibited superior performance compared to the control group without feedback for cognitive reappraisal of emotional stimuli characterized by low positive valence, but not for stimuli with high negative valence. This may be attributed to the relative ease of regulating high-negative emotions, where the contribution of neurofeedback signals is minimal; whereas regulating low-positive emotional states is more challenging, and subjects can leverage neurofeedback signal to optimize cognitive reappraisal strategies, thereby improving regulation performance. This result underscores the importance of considering regulation direction when implementing neurofeedback-enhanced cognitive reappraisal training.

Neural correlates for neurofeedback-guided cognitive reappraisal

While emotions are inherently difficult to describe directly, they can be detected through physiological indices, such as EEG. Numerous researchers focusing on emotion recognition have employed EEG-based methods because signals acquired from accessible EEG devices can provide information-rich data with high temporal resolution. Based on existing literature, emotions can be analyzed by utilizing distinguishable features of EEG signals (Liu et al. 2021; Rahman et al. 2021). We conducted an analysis of selected

EEG features for emotion classification at the group level, identifying that wavelet energy features and DE features play a particularly significant role in emotion classification. Specifically, the frequency components of beta and gamma of the anterior frontal, bilateral temporal, posterior parietal, and occipital regions were found to be crucial. Consistent findings have been reported in previous studies examining EEG features in emotion recognition (Wang et al. 2014; Zheng et al. 2019). This can be attributed to the notion that higher frequency brain activities are indicative of emotional and cognitive processes (Ray and Cole 1985; Luther et al. 2022). Additionally, the relationship between emotion recognition and frontal regions was elucidated in the studies of (Lin et al. 2010). Existing literature also supports the involvement of the temporal cortex in the processing of visual materials with emotional content (Meletti et al. 2006), with the temporal lobe being suggested as informative for detecting pleasant affect (Kortelainen et al. 2015). Regarding time-domain features, discriminating features of ERP component P300 have been identified over central and posterior regions. Studies have validated the relationship between the amplitude of P300 and the valence of emotional stimuli (Zhu et al. 2015). Hjorth parameter features, the complexity and mobility features, are also among the top selected features. This findings aligns with previous studies that claim the importance of Hjorth parameters in EEG classification tasks (Cecchin et al. 2010; Türk; et al. 2017).

Neurofeedback is provided to a subject based on his/her brain activity to facilitate self-regulation of brain function. Consequently, alternations in selected features of emotion decoding could reflect the neural plasticity changes induced by emotion regulation. Overall, both groups of subjects exhibited relatively consistent changes in EEG features, primarily located in the frontal and parietal regions. For instance, a decrease in P300 amplitude was observed for both groups. Because differences in valence dimensions can modulate ERP outcomes (Bernat et al. 2001), this result is indicative of a more positive perception. Furthermore, we observed alternations in high-frequency band oscillations

during emotion regulation, which are closely linked to emotions. For example, one study observed stronger responses in the Gamma band for pleasant pictures compared to other stimuli (Boucher et al. 2015). Additionally, these high-frequency band activities serve as indicators of emotion integration characteristics, particularly in the cognitive control of emotions (Tang et al. 2011).

Significant correlations were observed between neurofeedback training performance and changes in EEG activities. Specifically, in the neurofeedback group, notable correlations were found in the high-frequency Wavelet energy of the left temporal region, whereas in the control group, correlations were observed in the Hjorth mobility of the anterior frontal region. Cognitive reappraisal involves the process of reinterpreting an emotional stimulus to alter the emotional response and achieve emotion regulation. Therefore, language and memory processes are also crucial for the successful implementation of cognitive reappraisal. The frontal lobes and emotions are closely linked and serve as key brain regions for emotion regulation. Activity in the left frontal lobe is associated with approach motivation and positive emotions, while activity in the right frontal lobe is linked to avoidance motivation and negative emotions (Baehr et al. 1997). Additionally, the temporal lobe plays a crucial role in the semantic system (Binder et al. 2009) and facilitates the storage and retrieval of relevant information (Jackson 2021). Numerous studies have consistently reported activation of temporal lobe areas of the brain during cognitive reappraisal (Goldin et al. 2008; Kanske et al. 2011; Dörfel et al. 2014). Therefore, the observed differences between the two groups may stem from the adoption of different cognitive reappraisal strategies during neurofeedback training.

The present study has several limitations and the findings require further validation. First, a single-session neurofeedback training design was applied and the training performance was evaluated within a short period. Previous research in this field has predominantly focused on training performance across multiple training sessions, examining both the self-regulation of EEG activities and the long-term effects on behavioral performance (Bu et al. 2019; Huang et al. 2021). Secondly, this study solely assessed learning performance during neurofeedback training, lacking an indicator for immediate behavioral assessment of the short-term effects of emotion regulation. Further investigation is warranted to illustrate the complex relationship among the inter-individual differences in the decoded EEG neurofeedback training protocol (such as classification accuracy), neurofeedback learning performance, and behavioral training efficiency. Thirdly, future research also needs to explore the application of more sophisticated machine learning algorithms to optimize real-time decoding of emotional

states, to construct more accurate feedback signals (Jafari et al. 2023; Özçelik and Altan 2023a, b). Fourthly, the differences in emotional stimulus materials may influence the effectiveness of emotion regulation by affecting the induced emotional states. Currently, video-based emotional paradigms are gaining increasing attention, and future research could optimize training outcomes by substituting videos for images (Meer et al. 2020). Fifthly, given the known gender differences in emotion regulation (Nolen-Hoeksema and Aldao 2011), future studies with larger sample sizes could report data analysis results separately for male and female subjects. Lastly, it remains unclear how this decoded EEG neurofeedback-guided cognitive reappraisal training paradigm can be applied to patient groups, such as those with major depression. A potential line of research is to further verify the ability of this training paradigm in patients with mental disorders.

Conclusions

In the current study, we developed a novel cognitive reappraisal emotion training protocol guided by decoded EEG neurofeedback and made a comparison of training performance between different regulation directions. A single session of neurofeedback training facilitated enhanced positive emotions for emotional stimuli with low positive valence. Furthermore, the alternations in emotional states were associated with neural plasticity changes induced by cognitive reappraisal for emotion regulation. Specifically, we identified that wavelet energy features and DE features in the high-frequency band are particularly significant in emotion classification. We also observed alternations in high-frequency band oscillations during emotion regulation, and significant correlations were observed between these alternations and the performance of neurofeedback training. These findings indicate the beneficial effects of the proposed EEG neurofeedback protocol and offer a new neuromodulation technique that could be applied in patients with clinical mood disorders.

Acknowledgements We want to thank all the subjects who participated in this study.

Author contributions Conception and design of this study: L.L., Z.Z. and Z.L.; Acquisition of data: X.G. and G.H.; Methodology: L.Z. and F.W.; Formal analysis of data: L.L. and X.G.; Drafting of the article: L.L. and X.G.; Review & Editing: X.H., J.W. and D.N.; Critical revision of the article: Z.L. and Z.Z.; All authors reviewed the findings and interpretation and approved its final, submitted form.

Funding This work was supported in part by the National Natural Science Foundation of China (No. 82272114 and 62276169), in part by the Shenzhen Science and Technology Program (No. RKX20220705152815035, KCXFZ20201221173613036, and

2022SHIBS0003), in part by the Shenzhen University-Lingnan University Joint Research Programme, and in part by the Medical-Engineering Interdisciplinary Research Foundation of Shenzhen University.

Data availability The data and code used to support the findings of this study are available from the corresponding author upon request.

Declarations

Conflict of interest The authors have no competing interests to declare that are relevant to the content of this article.

Ethics approval The study protocol was approved by the Human Research Ethics Committee of Shenzhen University (No. 2019017). Informed, written consent was obtained from all participants and their caregivers prior to participation in the study.

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